

KUNAL MATHUR

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SUMMARY

Full Stack Developer and Software Engineer with production experience building and shipping scalable systems across **web applications, distributed architectures, machine learning pipelines, and blockchain/Web3 platforms**. Proficient in **React, Node.js, FastAPI, Python, Java, and PostgreSQL**. Experienced delivering features in **Agile/Scrum** environments within high-throughput financial domains, designing **ETL pipelines** and **ML inference systems**, and building **3D interactive** and **real-time** web experiences. Strong foundation in **system design, microservices, CI/CD, and cloud deployment**.

EDUCATION

Vellore Institute of Technology <i>B.Tech in Computer Science & Engineering (Specialization: Blockchain) CGPA: 7.5</i>	2022 – Present Vellore, Tamil Nadu
Turning Point School <i>CBSE Class XII 79%</i>	2020 – 2022 Ajmer, Rajasthan
Mayoor School <i>CBSE Class X 90%</i>	2018 – 2020 Ajmer, Rajasthan

EXPERIENCE

YES Securities (Yes Bank) <i>Full Stack Developer Intern</i> - Developing and shipping production-grade full-stack features for the OMNI trading platform using React, Node.js, and RESTful APIs; delivering iteratively within Agile/Scrum sprint cycles. - Engineered scalable backend microservices and optimized API performance across a distributed financial system, directly improving reliability and reducing response latency in a high-throughput environment. - Led PostgreSQL database schema optimization and contributed to CI/CD pipeline improvements using Git-based version-controlled workflows and automated deployments. - Participated in system design discussions covering service decomposition, API contracts, data flow architecture, and fault-tolerant patterns for a live financial platform.	Jan 2026 – Jun 2026 Bengaluru, India
Team Levitate Hyperloop (Technical Team) <i>Senior Engineer — AI & Software Systems</i> - Led end-to-end design, development, and cloud deployment of levitatehyperloop.com — a fully responsive, SEO-optimized website featuring real-time 3D simulations in Three.js and React. - Architected backend services, embedded system interfaces, and real-time data pipelines within the pod's circuit and software architecture; applied OOP design principles throughout the full stack. - Managed full DevOps lifecycle — deployment, performance optimization, and version control for a globally accessible production system with real-time hardware data integration.	2024 – 2025 Vellore, India
E-Cell (Entrepreneurship Cell) <i>Junior Core Member</i> Co-organized E-Summit and Futurepreneurs events; coordinated cross-functional planning, logistics, and stakeholder engagement across campus-wide entrepreneurship initiatives.	2023 – 2024 Vellore, India

PROJECTS

Real-Time Eye-Tracking Inference System - Building a real-time gaze estimation system in Python using computer vision and deep learning; implemented Kalman smoothing, temporal voting ensemble, and calibration improvements for robust inference without model retraining. - Conducting webcam domain adaptation research to bridge the gap between training data and real-world USB webcam deployment conditions.	2026 – Present
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Digital Chaos Lab — Real-Time Distributed Systems Simulator 2026

- Built a full-stack **distributed systems simulator** using **React (ReactFlow)**, **FastAPI**, and **WebSockets**; modeled failure scenarios via **M/M/1 queueing theory**, chaos engineering, and probabilistic fault injection (node crashes, latency spikes, traffic surges).
- Designed a **live metrics dashboard** for real-time observation of system state, throughput, and recovery patterns under varied fault conditions.

Hybrid Stock Price Prediction System | 2026

- Architected a **hybrid ML ensemble** — **LSTM + LightGBM** with **FinBERT NLP sentiment analysis** and technical indicators — for real-time stock forecasting; built automated **ETL and feature engineering pipelines** with model retraining workflows.
- Delivered end-to-end as a live web app at ysil.vercel.app with a **React framework** and real-time data visualization.

Fertilizer Subsidy Distribution Platform — Blockchain / Web3 2025

- Built a **DApp** for tamper-proof subsidy allocation using **Ethereum smart contracts (Solidity)** and **Web3.js**; **React + MetaMask** frontend with on-chain redemption logic and event-driven audit trails.

Multi-Source Job Aggregation & Ranking System 2025

- Built an **ETL pipeline** using web scraping to aggregate, clean, and normalize job listings; implemented a **weighted scoring model** ranking opportunities by skill match, experience, and domain relevance.

Official Website — Team Levitate Hyperloop | 2024

- Designed and deployed a **3D interactive, fully responsive** website in **Three.js and React** with performance-optimized rendering — live at levitatehyperloop.com.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript (ES6+), Solidity

Frontend: React.js, Three.js, ReactFlow, HTML5, CSS3, Responsive UI Design

Backend: Node.js, FastAPI, Express.js, RESTful APIs, WebSocket, Microservices, System Design

Databases: PostgreSQL, MongoDB, Firebase

ML / AI: TensorFlow, Scikit-learn, LSTM, LightGBM, NLP (FinBERT), Deep Learning, Computer Vision, Kalman Filtering, Feature Engineering, ETL Pipelines, Data Analysis, Web Scraping

Blockchain / Web3: Ethereum, Solidity, Smart Contracts, Web3.js, DApps, Hyperledger Fabric

Tools & Cloud: Git, GitHub, AWS, Vercel, Figma, Blender

Concepts: Distributed Systems, Chaos Engineering, CI/CD, Agile/Scrum, SDLC, OOP

ACHIEVEMENTS

Winner — Global Hyperloop Competition (Asia's First Hyperloop Competition, IIT Madras) 2025
Secured **six awards** across engineering, design, and technical presentation — representing VIT Vellore.

National Semi-Finalist — Flipkart GRiD 7.0 2025
Advanced to **national semi-finals** in a highly competitive real-world software engineering challenge.

CERTIFICATIONS

Blockchain Developer Certification — IBM Career Education 2025
Blockchain fundamentals, smart contract development, Hyperledger Fabric, and DApp architecture.

Full Stack Web Development (MERN) — Apna College 2024
Built and deployed multiple production full-stack web applications end-to-end.